abundance with a mass of 62.9296 u and of a second isotope of 30.83% abundance and 64.9278 u. Atomic masses and abundances of some isotopes are given in Table 3.1.

3.3. DETERMINATION OF ISOTOPIC MASSES AND ABUNDANCES

3.3.1. The mass spectrometer

The masses and relative abundances of different isotopes occurring naturally in an element can be determined with great exactness using the same technique J. J. Thomson employed to demonstrate the presence of isotopes in neon. The instrument used for this purpose is known as a mass spectrometer. The principles of the electromagnetic *mass spectrometer* are shown in Figs. 3.1 and 3.2.

Let us first consider the movement of an ion in electric and magnetic fields, as shown in Fig. 3.1. The ion of mass m is assumed to have a charge q (coulomb), which is an integer (z) multiple of the elementary charge **e** ($1.602\ 177 \times 10^{-19}$ C): $q = z\mathbf{e}$. If it is accelerated from velocity zero to v (m s^{-1}) by an electric potential V (volts), it acquires a kinetic energy E_{kin} corresponding to joule (or Newton meter).

$$E_{kin} = \tfrac{1}{2}mv^2 = qV \tag{3.6}$$

If q is given in units of the elementary charge, the kinetic energy is in units of *electron volts* (eV). For transformation to other energy units, see Appendix IV.

Figure 3.1 shows the deviations of a positive ion in an electric field U (newton/coulomb) directed upwards, and a magnetic field B (tesla) directed into the page. The force F (newton) acting on the ion due to the electric field is

$$F_e = qU \tag{3.7}$$

If the electric force would dominate, the ion would hit the screen at point P_1. For the magnetic field only, the force is

$$F_m = qvB \tag{3.8}$$

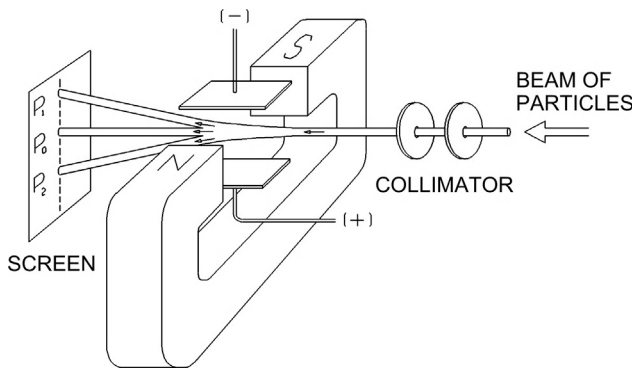


Figure 3.1 Movement of positive ions in electric and magnetic fields.

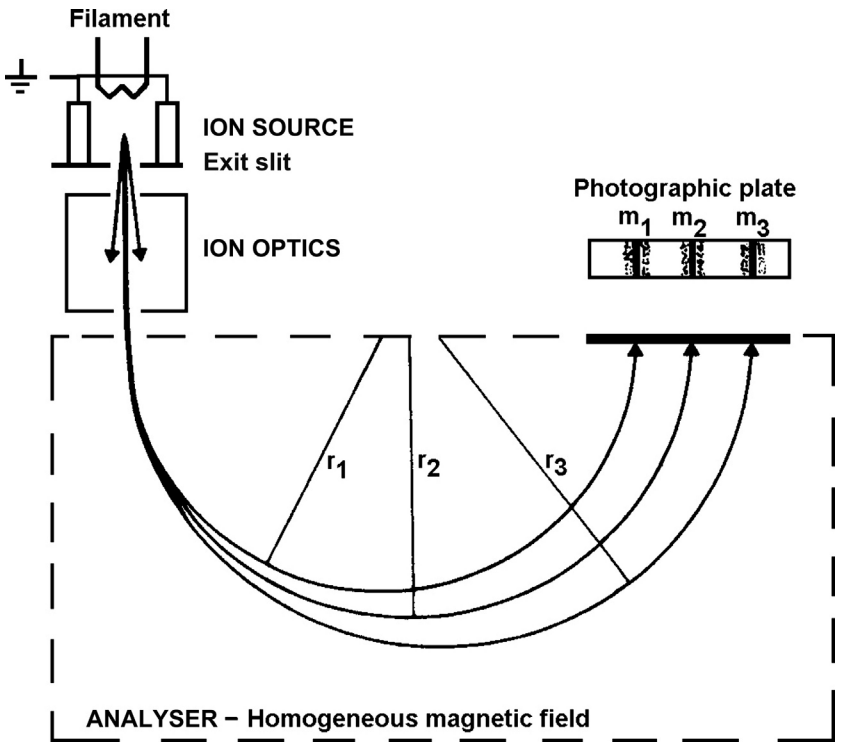


Figure 3.2 The principle of the mass spectro*graph* (*-meter*, if the photographic plate is replaced by an ion current meter).

and if this force would dominate the ion would hit the screen at point P_2. If the forces balance each other, i.e. when $F_e = F_m$, the ion hits the screen at point P_0 with a velocity given by

$$v = U/B \tag{3.9}$$

In either the fields (F_e or F_m) the deviation is counteracted by the centrifugal force F_c, where

$$F_c = mv^2/r \tag{3.10}$$

and r is the radius of curvature. In the magnetic field only, the balance between F_m and F_c leads to

$$q/m = v/Br \tag{3.11}$$

where q/m is denoted as the *specific charge* of the ion.

In the mass spectrometer gaseous ions are produced in an *ion source*, e.g. by electron bombardment of the gas produced after e.g. heating the substance in a furnace (Fig. 3.2) or by electric discharge, etc. If positive ions are to be investigated, the ion source is given

a high positive potential relative to the exit slits. This results in the ions being accelerated away from the source and into the ion optic system. The purpose of the ion optic system is to produce ions of exact direction and of constant velocity, which is achieved through the use of electrostatic and magnetic fields as described; cf. (3.9).

The spectrometer commonly consists of a homogeneous magnetic field which forces the ions to move in circular paths according to (3.10). Combining (3.6) and (3.11) gives

$$m = q\, r^2\, B^2/2V \tag{3.12}$$

where V is the ion acceleration potential. Mass spectrometers are usually designed so that of the three variables V, B, or r, two are constant and the third variable, which allows ions of different q/m value to be focused on the detector. The minimum value of q/m is always $\mathbf{e}/m$ because singly charged ions of the atomic or molecular species are almost always present. In order to avoid collisions between ions and gaseous atoms, which would cause scattering of the ion beam, the system is evacuated. The detector is some charge collecting device (Ch. 9).

A common type of mass spectrometer (A. O. Nier 1940) uses a fixed magnetic field and a fixed radius of curvature for the ion beam. If the acceleration potential V is varied so that the masses m_1 and m_2 alternately are registered by the detector, producing ion currents I_1 and I_2, respectively, the abundance of each isotope can be calculated from the ratios $x_1 = I_1/(I_1 + I_2)$ and $x_2 = 1 - x_1$ when only two isotopes are present. The resolution of modern mass spectrometers can be extremely high, as indicated by the values in Table 3.1.

3.3.2. Applications

For several decades, mass spectrometers were used primarily to determine atomic masses and isotopic ratios. Now they are applied to a large variety of chemical problems and low resolution mass spectrometers are used for routine chemical analysis. For example, a modern mass spectrometer can easily distinguish between species such as $^{12}CH_4^+$ and $^{16}O^+$, having a mass difference of 0.03686 u. The resolution (R) of a mass spectrometer is defined as:

$$R = M/\Delta M \tag{3.13}$$

where M is the atomic mass and ΔM is the minimum peak separation. There are, however, two different methods to define the minimum peak separation:

The peak width definition: ΔM is given based on the width of the peak at a specific peak height, for example 5%. This definition is also used in e.g. gamma spectroscopy (Ch. 9)

The valley definition: ΔM is given by the closest spacing between two peaks of equal intensity where the intensity of the valley is less than a specific fraction of the peak height. Typical values are tenths of percent. For comparison, the value obtained from a 5% peak width is roughly equivalent to a 10% valley.

Some uses of mass spectrometry of interest to chemists involved in nuclear science are:

(a) *Molecular weight determination* can be made by mass spectrometry if gaseous ions can be produced with M/q values not exceeding about 400. This method is of great importance in radiation chemistry (Ch. 8) where often a large number of products are produced which may be quite difficult to identify by other means and in particular for the analysis of organic compounds.

(b) The *study of chemical reactions* directly in the gas phase by mass spectrometry is possible. Using an ion source in which molecules are bombarded by a stream of low energy ($\leq$100 eV) electrons, ionization and dissociation reactions can be studied, e.g.

$$\begin{array}{l} C_8H_{18} + e^- \rightarrow C_8H_{18}^+ + 2e^- \\ \qquad\qquad\quad \big\downarrow \\ \qquad\qquad\quad \rightarrow C_4H_9^+ + C_3H_6^+ + CH_3\cdot + e^- \end{array} \tag{3.14}$$

This technique has practical application e.g. in the petroleum industry for determining the composition of distillation and cracking products.

(c) *Isotopic dilution* is a technique for determination of the number of atoms of an element (or isotope) in a composite sample (e.g. rock or biota), from which is difficult to recover the element reproducibly and determine it quantitatively. The technique is simple to use for radioactive nuclides, as described in §18.3.2, but more complicated and time consuming for stable nuclides. However, the high precision of mass spectrometry makes the latter alternative more accurate, and the technique is widely used for geologic dating.

Suppose we have a *sample* with *unknown* concentration of a certain element, consisting of the two isotopes 1 and 2, and want to determine the number of atoms, N_2, of isotope 2 in the sample (which, of course must be of known weight). The number of atoms of each isotope is N_1 and N_2, thus $N = N_1 + N_2$, in unit sample weight. By mass spectrometry we determine the isotope ratio $\zeta_N = N_1/N_2$.

In isotope dilution analysis the isotope ratio in a sample to be measured is altered by adding a known amount of a standard (spike) with a well known isotopic composition that differs from the isotopic composition in the sample. By measuring the isotopic composition of this mixture called blend, the amount of the analyte in the sample can be calculated. Of course, the isotopic composition of the sample has also to be known! Below, s refers to sample, sp to spike and B to blend. Moreover we consider only the two isotopes a and b. A mass balance for the two components 1 and 2 gives first for the number of moles of component a in the blend:

$$n_{1.B} = n_{1.s} + n_{1.sp} \tag{3.15}$$

And for component b in the same way

$$n_{2.B} = n_{2.s} + n_{2.sp} \tag{3.16}$$

Combine these two equations (division), and we'll get

$$\frac{n_{2.B}}{n_{1.B}} = \xi_B = \frac{n_{2.s} + n_{2.sp}}{n_{1.s} + n_{1.sp}} \tag{3.17}$$

Where ξ_B is the isotope ratio (mole ratio) of component 1 and 2 in the blend. We know that the number of moles of isotope j in e.g. the blend is $n_{j.B} = f_{j.B} \cdot n_B$ ($n_B = n_{j.B} + n_{k.B}$) and that the abundance $f_{j.B} = \xi_{j.B}/\Sigma\xi_{j.B}$. From this we get the isotope ratio of 1 and 2 in the blend

$$\xi_B = \frac{\frac{\xi_{2.s}}{\Sigma\xi_s} n_s + \frac{\xi_{2.sp}}{\Sigma\xi_{sp}} n_{sp}}{\frac{1}{\Sigma\xi_s} n_s + \frac{1}{\Sigma\xi_{sp}} n_{sp}} \tag{3.18}$$

Note that here $\xi_{1.B} = 1$! After simplification we get

$$n_s = n_{sp} \frac{\xi_{sp} - \xi_B}{\xi_B - \xi_s} \cdot \frac{\Sigma\xi_s}{\Sigma\xi_{sp}} \tag{3.19}$$

But $n = C \cdot m$, so we get

$$C_s = C_{sp} \frac{m_{sp}}{m_s} \cdot \frac{\xi_{sp} - \xi_B}{\xi_B - \xi_s} \cdot \frac{\Sigma\xi_s}{\Sigma\xi_{sp}} \tag{3.20}$$

If the sample can be considered to have a natural isotopic composition of the analyte (or any other well known composition), we only have to measure the blend ratios. Using mass spectrometry we then get

$$C_s = C_{sp} \frac{m_{sp}}{m_s} \cdot \frac{\xi_{sp} - K \cdot r_B}{K \cdot r_B - \xi_s} \cdot \frac{\Sigma\xi_s}{\Sigma\xi_{sp}} \tag{3.21}$$

where r_i is an observed ratio and K a correction factor due to mass fraction within the mass spectrometer, i.e. $K = \xi_i/r_i$. Mass fractionation may depend on several processes depending on e.g. the ion source. The light isotope will deviate from the ion beam to a greater extent than the heavy one due to electrostatic repulsion. In ion sources where diffusion processes plays a role there will also be a mass fractionation since the light isotope will have a higher diffusivity compared to the heavy one.

Thus from 3 mass spectrometrically determined isotope ratios (ζ_s, ζ_{sp} and ζ_B) and known standard amount $P = P_1 + P_2$, the unknown number of atoms N_2 in the sample is determined. The precision of this technique is largest when sample and

standard are added in such proportions that $N \sim P$. Examples of the use of this technique are given in §13.8 on geologic dating.

In the case of using a radioactive isotope for the isotopic dilution equation 3.20 simplifies to

$$W_x = W_0\left(\frac{S_0}{S_2} - 1\right) = W_0\left(\frac{A_0 W_2}{A_2 W_0} - 1\right) \tag{3.22}$$

where W_x is the weight of the unknown, W_0 is the weight of the spike added, A_0 its activity, and S_0 its specific activity ($S_0 = A_0/W_0$). Similarly W_2 is the weight of the analyte isolated, A_2 its activity, and S_2 its specific activity ($S_2 = A_2/W_2$). When only the ratio S_0/S_2 is used, the activities A_0 and A_2 can be replaced by the measured radioactivities R_0 and R_2 in case that the detection efficiency ψ is the same.

Because the variables in these equations often need multiple indexing (e.g. element, isotope and source) the amount of indexing can be reduced by using *italic* element symbols to refer to the specific radioactivity, or concentration, of that nuclide in the sample; e.g. ^{238}U means S_{238U} (radioactivity), or N_{238U} (atoms) per unit volume or weight of the *sample*. Extensive use of this formalism will be found in §18.3.2***ff***[1].

(d) *Analysis of gas purity* (e.g. in plants for separation of ^{235}U and ^{238}U) is done conveniently by mass spectrometry. Not only are the ratios of the uranium isotopes determined, but also the air and water vapour which leaks into the system can be measured. This produces such ions as O_2^+, N_2^+, CO_2^+, and HF^+, which can be measured easily because mass spectrometry detects the presence of impurities in parts per million (ppm) concentration.

(e) *Instrumental chemical analysis* can be done by using mass spectrometers having ion sources of the inductively coupled plasma type (ICP-MS systems), sometimes as on-line detectors for various chromatographic techniques. Due to the high resolution and sensitivity of this technique it is widely used in analysis of pollutants in the environment.

(f) *Measurement of products from nuclear reactions* can be made with special mass spectrometers on line to the irradiated target. With suitable arrangements both A and Z of the recoiling reaction products can be determined.

It needs to be noted here that although ppm or similar mass units are used in mass spectroscopy these units are not optimal for use in chemical analysis. Here there might be a need to assess whether there is enough of a specific ligand to reach equilibrium or to determine of a solution is super saturated with respect to an element. Thus for chemical use mass units are preferably converted to amount units like number of moles per volume.

[1] The Student making notes may have difficulties in distinguishing between italics and normal letters and is therefore recommended to adhere to the use of multiple indexing.

3.4. ISOTOPIC RATIOS IN NATURE

Mass spectrometric investigations of geologic material has shown that isotopic ratios vary slightly in nature with the largest variations observed for the heaviest and lightest elements. Hence the atomic weight of many elements is a little uncertain, which is indicated be giving fewer decimals than for their isotopes.

For the heaviest elements the reason for the variation in the ratio of isotopes in an element can be related directly to the consequence of naturally occurring radioactive decay. For example, in a thorium mineral the decay sequence terminates in the formation of an isotope of lead with mass number 208. By contrast, in a uranium ore, in which ^{238}U is the most abundant uranium isotope, the primary end product of the decay is an isotope of lead with mass number 206 (see Fig. 1.2). This means that thorium minerals and uranium minerals contain lead with different isotopic composition. In fact, one of the best confirmations of the existence of the radioactive decay series came from the proof that lead had different net atomic weights in different minerals.

Lead from different ores will contain slightly different isotopic abundances due to the fact that all long natural radioactive decay series end up in differet isotopes of lead An unusual use of this fact was made by Andrasko et al. to identify smears and fragments from lead bullets used in a homicide case, so that the suspect could be bound to the case, as the isotopic composition of lead bullets can be identified not only by manufacturer but also by manufacturing date.

The isotopic ratios for the lightest elements depend on in which natural material they are found. The ratio for $^{7}Li/^{6}Li$ varies from 12.20 to 12.48, while that of $^{12}C/^{13}C$ varies from 89.3 to 93.5. The isotopic ratio $^{18}O/^{16}O$ varies by almost 5% as shown in Fig. 3.3. However, since natural oxygen contains only 0.2% ^{18}O, even a 5% variation in the isotopic ratio has very little influence on the atomic weight of oxygen. The variation of natural isotopic ratios for boron and chlorine causes the uncertainties in their abundances shown in Table 3.

3.5. PHYSICOCHEMICAL DIFFERENCES FOR ISOTOPES[2]

Although the isotopic variations in the heaviest elements can be attributed to the consequences of radioactive decay, the variations observed in lighter elements are attributable to chemical behaviour. The rates and equilibria of chemical reactions depend on the masses of the atoms involved in the reactions as is explained in §3.6 and §3.7. As a consequence, isotopes may be expected to have somewhat different quantitative physicochemical values (the *isotope effect*). This effect is naturally more predominant for lighter elements than heavier since for the lighter elements the change in mass by adding a neutron

[2] §§3.5-3.7 outline the scientific basis for isotope effects and isotope separation. This is not essential for the Radiochemistry part of this book.